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Editor-in-Chief
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Dear Editor,

We are pleased to submit our manuscript titled "A Comparative Study of Convolutional and Transformer Architectures Under Adversarial Perturbations: Gradient Characteristics and Transferability Analysis" for consideration as a Regular Article in *IEEE Access*.

This paper provides a behavioral characterization of how convolutional neural networks (CNNs) and Vision Transformers (ViTs) differ under adversarial perturbations. Rather than merely comparing robustness scores, we examine *how* these architectures behave differently under attack, an increasingly important question as both architecture families are deployed in safety-critical systems. Using CIFAR-10 with adversarially trained ResNet-18 and ViT-Tiny models evaluated on the full test set under AutoAttack ($\epsilon = 8/255$), we present four key findings:

- **Robustness comparison with paired statistics:** The CNN attains a statistically significant but modest 2.8-point advantage under full-test AutoAttack (35.74% vs 32.94%; McNemar paired test on 10,000 samples, $p < 10^{-11}$).
- **A methodological caution for transfer studies:** Under the widely used raw transfer metric our models exhibit an apparent 8.3-point cross-architecture asymmetry; under a clean-accuracy-controlled (conditioned) fooling-rate metric the asymmetry vanishes entirely (20.95% vs 20.32%, overlapping confidence intervals). We quantify, in a controlled setting, how large a spurious asymmetry the raw metric can manufacture when targets differ in clean accuracy.
- **Gradient characteristics as behavioral correlates:** Under scale-invariant sparsity measures, CNN gradients are modestly more concentrated (Hoyer 0.474 vs 0.449)—a difference that survives a native-resolution ViT control—while cross-sample gradient alignment is low for both models yet 1.36× higher for the ViT (0.052 vs 0.038).
- **Layer-wise feature degradation profile:** Adversarially trained ViT representations under attack exhibit an early-drop, mid-network-plateau profile: clean–adversarial cosine similarity falls from 0.994 to a minimum of 0.955 by mid-network and then plateaus, while feature norms change by less than 1.5%.

We believe this work is well suited for *IEEE Access* because of the journal's interdisciplinary readership spanning computer vision, machine learning security, and deep learning systems. Our analysis bridges adversarial robustness and architectural understanding, topics that are highly relevant to the diverse IEEE community. The open-access nature of *IEEE Access* will also maximize the impact of these findings for researchers and practitioners working on secure AI deployment.

**Declarations:**

- This manuscript is original work and has not been published elsewhere.
- The manuscript is not under consideration by any other journal.
- All authors have reviewed and approved the final manuscript.
- All experiments use the publicly available CIFAR-10 dataset and standard evaluation protocols.
- The authors declare no conflict of interest.
- Code and trained models will be made publicly available upon acceptance.

Thank you for considering our manuscript. We look forward to your editorial decision.

Sincerely,

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